

# **Chi-Square Testing and ANOVA Applied Analysis of Categorical and Continuous Data**

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### **Introduction:**

The purpose of this report is to apply chi-square hypothesis testing and analysis of variance (ANOVA) techniques to a series of applied problems drawn from Bluman's Elementary Statistics, as well as two independent datasets on baseball and crops. The goal of the assignment is to demonstrate mastery of statistical procedures: (1) the chi-square goodness-of-fit test, which determines whether an observed distribution of a categorical variable matches a hypothesized distribution, (2) the chi-square test for independence and homogeneity which determines whether two categorical variables are related or whether proportions are consistent across populations, (3) one-way ANOVA which tests whether three or more population means differ, and (4) two-way ANOVA which extends the one-way test to evaluate the main effects of independent factors and their interaction on a continuous outcome. The textbook exercises cover blood type distribution in a hospital population, on-time airline performance, movie admissions by ethnicity and year, the rank distribution of woman across military branches, sodium content across food categories, sales for leading consumer product companies, per-pupil educational expenditures across US regions, and a two-factor greenhouse experiment on plant growth. Each of these problems is solved using the traditional (critical value) method of hypothesis testing, with the null and alternative hypotheses, critical value, test statistic, decision, and summary reported for every test. In addition to the textbook exercises, this report analyzes two supplementary datasets in R with the first being `baseball.csv` and containing 1,232 team season records from the Major League Baseball spanning 1962 – 2012 and the variables describing runs scored, wins, on-base percentage, slugging percentage, batting average, and playoff outcomes. Exploratory data analysis is performed on this dataset, followed by a chi-square goodness of fit test to determine whether the total number of wins is evenly distributed across decades. The second dataset

crop\_data.csv contains yield measurements from an agricultural experiment involving two levels of planting density and three fertilizer treatments across four blocks. A two-way ANOVA is used to evaluate whether fertilizer type, planting density, or their interaction significantly affects crop yield.

### Analysis:

#### Section 11-1

#### Question 6 Blood Types:

A medical researcher wants to determine whether the blood type distribution among 50 randomly sampled hospital patients matches the known general-population distribution (A=20%, B=28%, O=36%, AB=16%). The Hypotheses are  $H_0$ : the distribution of blood types in the hospital follows the general population distribution (this is the researcher's claim) and  $H_1$ : the distribution of blood types in the hospital does not follow the general population distribution. At  $\alpha = 0.10$ , with 4 categories and 3 degrees of freedom, the critical value from the chi-square distribution is 6.251.

| BloodType | Observed | ExpectedProportion | Expected |
|-----------|----------|--------------------|----------|
| A         | 12       | 0.2                | 10       |
| B         | 8        | 0.28               | 14       |
| O         | 24       | 0.36               | 18       |
| AB        | 6        | 0.16               | 8        |

The computed test statistic is chi-square = 5.471 which is less than the critical value of 6.251.

Since the test statistic does not fall in the critical (rejection) region, the decision is to fail to reject  $H_0$ . There is not sufficient evidence at  $\alpha = 0.10$  to conclude that the blood type distribution among hospital patients differs from that of the general population.

#### Question 8: On-Time Performance by Airlines

The Bureau of Transportation Statistics reports that nationally 70.8% of flights are on time, 8.2% are delayed by the National Aviation System, 9.0% arrive late for other reasons, and 12.0% are affected by weather or other conditions. A sample of 200 flights for a major airline showed 125 on time, 40 delayed due to the weather, 10 delayed by the National Aviation System, and the remaining 25 arriving late.  $H_0$  is the airline's on-time performance distribution is the same as the government's statistics and  $H_1$  is the airline's distribution differs from the government's statistics. At  $\alpha = 0.05$  with 3 degrees of freedom, the critical value is 7.815.

| Category     | Observed | ExpectedProportion | Expected |
|--------------|----------|--------------------|----------|
| OnTime       | 125      | 0.708              | 141.6    |
| NASDelay     | 10       | 0.082              | 16.4     |
| ArrivingLate | 25       | 0.09               | 18       |
| Other        | 40       | 0.12               | 24       |

The computed chi-square statistic is 17.833 which exceeds the critical value of 7.815 ( $p < 0.001$ ) so the decision is to reject  $H_0$ . There is sufficient evidence at  $\alpha = 0.05$  to conclude that this airline's on-time performance distribution differs significantly from the government's published statistics, supporting the claim of a difference.

## Section 11-2

### Question 8: Ethnicity and Movie Admissions:

This problem asks whether movie attendance (in thousands of admissions) is related to or dependent upon ethnicity across 2 years.  $H_0$  is movie attendance by year is independent of ethnicity and  $H_1$  is movie attendance by year is dependent upon ethnicity. With a 2 by 4 contingency table, degrees of freedom =  $(2-1)(4-1) = 3$  and at  $\alpha = 0.05$  the critical value is 7.815. The chi-square test statistic is 60.144 ( $p < 0.001$ ) far exceeding the critical value of 7.815. The decision is to reject  $H_0$ .

| Year | Caucasian | Hispanic | African American | Other |
|------|-----------|----------|------------------|-------|
| 2013 | 724       | 335      | 174              | 107   |
| 2014 | 370       | 292      | 152              | 140   |

There is sufficient evidence at  $\alpha = 0.05$  to conclude that movie attendance by year is dependent upon or related to ethnicity, which supports the claim.

#### Question 10 Women in the Military:

This problem tests whether a relationship exists between military rank (officer or enlisted) and branch of service (Army, Navy, Marine Corps, Air Force) for women in the US armed forces.  $H_0$  is that rank is independent of branch of service and  $H_1$  is a relationship exists between rank and branch which is the claim. With a 4 by 2 table, degrees of freedom  $(4-1)(2-1) = 3$ , and the critical value at  $\alpha = 0.05$  is 7.815

| Branch       | Officers | Enlisted |
|--------------|----------|----------|
| Army         | 10791    | 62491    |
| Navy         | 7816     | 42750    |
| Marine Corps | 932      | 9525     |
| Air Force    | 11819    | 54344    |

The computed chi-square statistic is 654.272 ( $p < 0.001$ ) which is well above the critical value of 7.815 so the decision is to reject  $H_0$ . There is sufficient evidence at  $\alpha = 0.05$  to conclude that a relationship exists between rank and branch of service among women in the military, supporting the claim

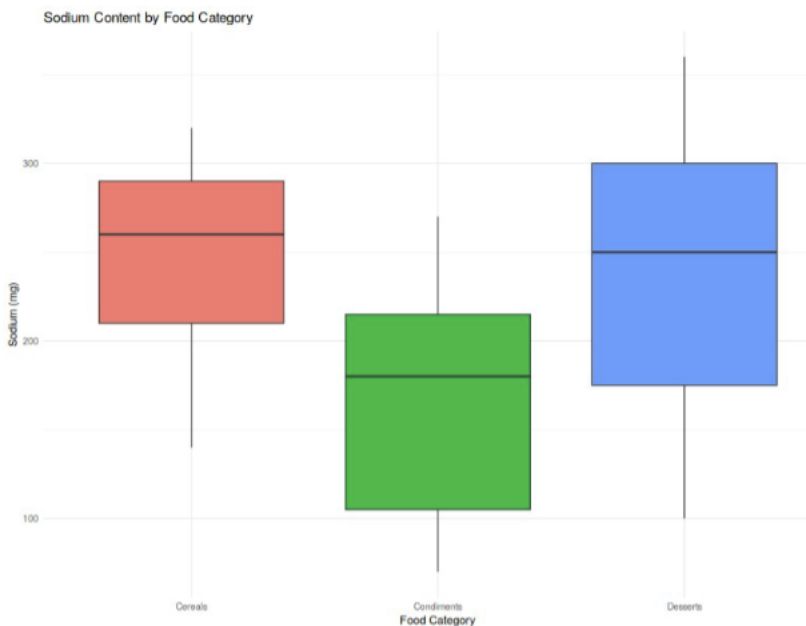
#### One-Way ANOVA: Sodium Contents of Foods

This model compares mean sodium content across three food categories which are condiments, cereals, and desserts. Since three independent group means are being compared, a one-way ANOVA is the appropriate test (rather than a two-sample test).  $H_0$  is that  $\mu_{\text{condiments}} = \mu_{\text{cereals}} = \mu_{\text{deserts}}$  so there is no difference in mean sodium content

and H1 is that at least one mean differs from the others. At  $\alpha = 0.05$ , with 2 and 19 degrees of freedom, the critical F value is 3.522.

| food       | n | mean   | sd    | min | max |
|------------|---|--------|-------|-----|-----|
| Cereals    | 7 | 245.71 | 62.68 | 140 | 320 |
| Condiments | 7 | 165.71 | 75.47 | 70  | 270 |
| Desserts   | 8 | 237.5  | 85.65 | 100 | 360 |

| Source    | Df | Sum Sq     | Mean Sq   | F value | Pr(>F) |
|-----------|----|------------|-----------|---------|--------|
| food      | 2  | 27543.506  | 13771.753 | 2.399   | 0.118  |
| Residuals | 19 | 109092.857 | 5741.729  | NA      | NA     |



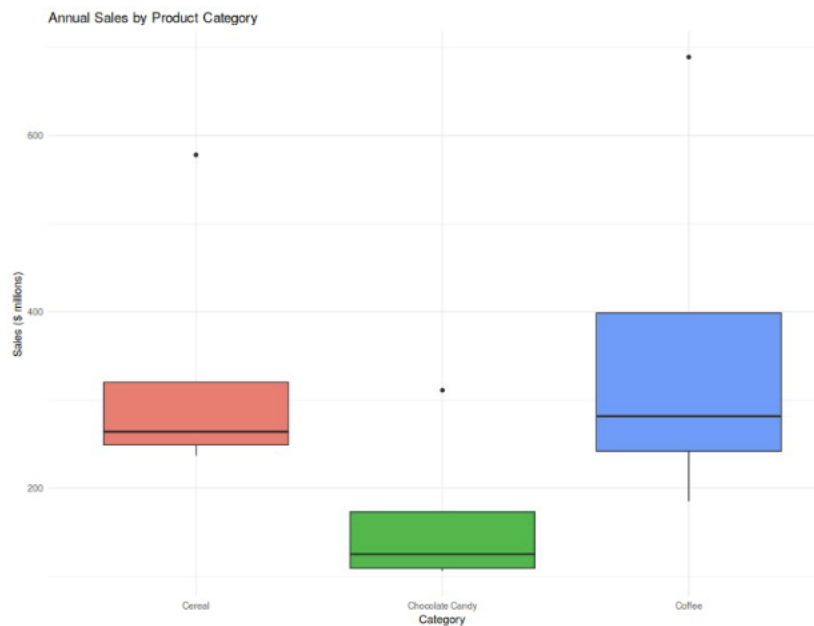
The computed F statistic is 2.399 ( $p=0.118$ ) which does not exceed the critical value of 3.522 so the decision is to fail to reject  $H_0$ . There is not sufficient evidence at  $\alpha = 0.05$  to conclude that a difference in mean sodium amounts exists among condiments, cereals, and desserts, even though the boxplot suggests cereals and desserts tend higher than condiments. Since the null hypothesis was not rejected, no Tukey or Scheffe test were performed.

Question 10: Sales for Leading Companies

This problem compares annual sales (in millions of dollars) among three product categories which are cereal, chocolate candy, and coffee.  $H_0: \mu_{\text{cereal}} = \mu_{\text{candy}} = \mu_{\text{coffee}}$  and  $H_1$  is that at least one mean differs which is the claim. At alpha 0.01, with 2 and 11 degrees of freedom, the critical F value is 7.206

| category        | n | mean   | sd     | min | max |
|-----------------|---|--------|--------|-----|-----|
| Cereal          | 5 | 329.6  | 142.45 | 237 | 578 |
| Chocolate Candy | 5 | 164.8  | 86.02  | 106 | 311 |
| Coffee          | 4 | 359.25 | 225.11 | 185 | 689 |

| Source    | Df | Sum Sq     | Mean Sq   | F value | Pr(>F) |
|-----------|----|------------|-----------|---------|--------|
| category  | 2  | 103769.607 | 51884.804 | 2.172   | 0.16   |
| Residuals | 11 | 262794.75  | 23890.432 | NA      | NA     |



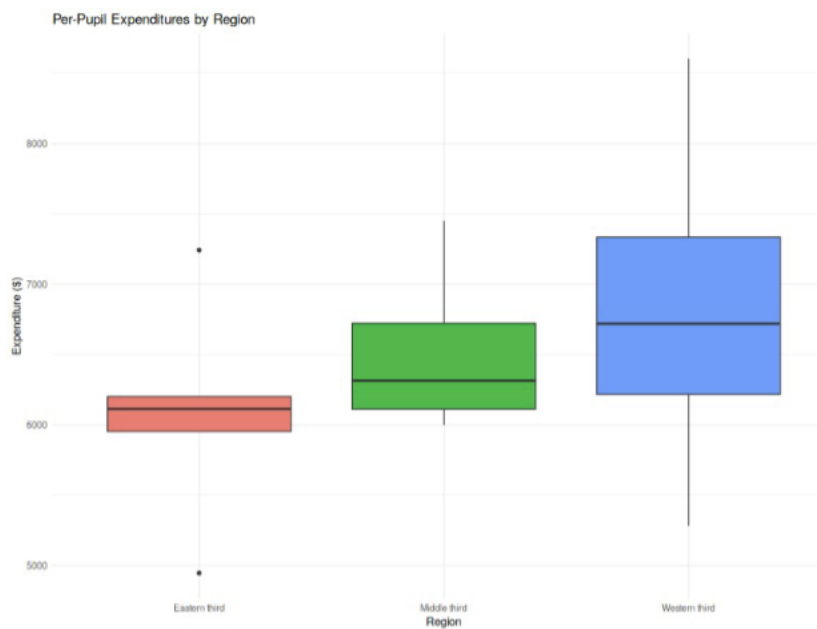
The computed F statistic is 2.172 ( $p = 0.160$ ) which does not exceed the critical value of 7.206 so the decision is to fail to reject  $H_0$ . There is not sufficient evidence at  $\alpha = 0.01$  to conclude that a significant difference in mean sales exists among cereal, chocolate candy, and coffee companies. Due to the fact that  $H_0$  was not reject, no Tukey or Scheffe post-hoc test was necessary.

### Question 12 Per-Pupil Expenditures:

This problem compares mean per-pupil expenditures across three US regions which are the eastern, middle, and western thirds of the country.  $H_0 = \mu_{\text{east}} = \mu_{\text{middle}} = \mu_{\text{west}}$  and  $H_1$  is that at least one regional mean differs. At  $\alpha = 0.05$ , with 2 and 10 degrees of freedom, the critical F value is 4.103.

| region        | n | mean    | sd     | min  | max  |
|---------------|---|---------|--------|------|------|
| Eastern third | 5 | 6091.4  | 817    | 4946 | 7243 |
| Middle third  | 4 | 6519.75 | 652.3  | 6000 | 7451 |
| Western third | 4 | 6831.5  | 1371.7 | 5282 | 8605 |

| Source    | Df | Sum Sq      | Mean Sq    | F value | Pr(>F) |
|-----------|----|-------------|------------|---------|--------|
| region    | 2  | 1244587.973 | 622293.987 | 0.649   | 0.543  |
| Residuals | 10 | 9591144.95  | 959114.495 | NA      | NA     |



The computed F statistic is 0.649 ( $p = 0.543$ ) which does not exceed the critical value of 4.103 and the decision is the fail to reject  $H_0$ . There is not sufficient evidence at  $\alpha = 0.05$  to conclude that per-pupil expenditures differ by region. The large within-group variability



particularly in the western third relative to the between-group variability explains the non-significant result and no post-hoc test was required

### Section 12-3

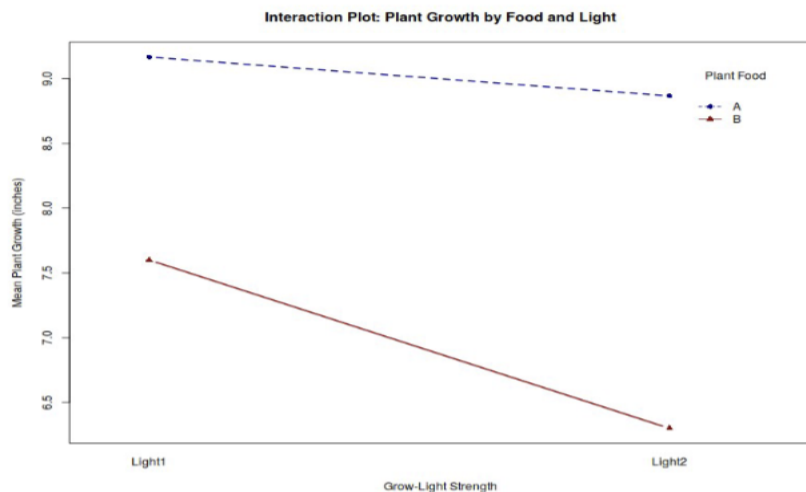
A gardening company tests whether grow-light strength (light 1 vs light 2) and plant food supplement (A vs. B) affect plant growth and whether the two factors interact. Three hypotheses are tested simultaneously where  $H_0$  (food) mean growth is the same for plant food A and B,  $H_0$  (light) mean growth is the same for light 1 and light 2,  $H_0$  (interaction) there is no interaction between plant food and grow-light. The corresponding alternative hypotheses state that a main effect or an interaction effect is present. At  $\alpha = 0.05$  with 1 and 8 degrees of freedom for each effect, the critical F value is 5.318

| food | light  | mean_growth |
|------|--------|-------------|
| A    | Light1 | 9.167       |
| A    | Light2 | 8.867       |
| B    | Light1 | 7.6         |
| B    | Light2 | 6.3         |

| Source     | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|------------|----|--------|---------|---------|--------|
| food       | 1  | 12.813 | 12.813  | 24.562  | 0.001  |
| light      | 1  | 1.92   | 1.92    | 3.681   | 0.091  |
| food:light | 1  | 0.75   | 0.75    | 1.438   | 0.265  |
| Residuals  | 8  | 4.173  | 0.522   | NA      | NA     |

The main effect of plant food is significant since  $F = 24.562$ ,  $p = 0.0011$ , which exceeds the critical value of 5.318 so  $H_0$  (food) is rejected. Plants receiving food A grew significantly more (mean approximately 9.02 inches) than plants receiving food B (mean approximately 6.95 inches). The main effect of grow-light is not significant with  $F = 3.681$ ,  $p = 0.091$ , which does not exceed 5.318 so  $H_0$  (light) is not rejected since there is not sufficient evidence that light

strength alone affects growth. The interaction effect is also not significant with  $F = 1.438$  and  $p = 0.265$ , so  $H_0$  (interaction) is not rejected.



As shown in the interaction plot, the lines for food A and food B are roughly parallel which is consistent with the absence of a statistically significant interaction, although food B shows a numerically larger drop in growth from light 1 to light 2 than food A does. Plant food is the dominant driver of growth in this experiment, while grow-light strength and the food-by-light interaction do not reach statistical significance at  $\alpha = 0.05$ .

#### Baseball Dataset:

The baseball.csv file was imported into R and contains 1,232 team season observations across 15 variables, covering the American League (616 observations) and the National League (616 observations) from 1962 through 2012. Key numeric variables include wins (W), runs scored (RS), runs allowed (RA) on-base percentage (OBP), slugging percentage (SLG), and batting average (BA). Descriptive statistics were exported directly from R rather than using the generic summary () function. On average, teams in the dataset won about 81 games per season ( $SD = 11.5$ ) with a range from 40 to 116 wins.

Runs scored and runs allowed are nearly identical on average (both approximately 715), which is expected because every run scored by one team is a run allowed by its opponent.

| Variable | n    | mean    | sd     | min   | max   |
|----------|------|---------|--------|-------|-------|
| BA       | 1232 | 0.259   | 0.013  | 0.214 | 0.294 |
| OBP      | 1232 | 0.326   | 0.015  | 0.277 | 0.373 |
| RA       | 1232 | 715.082 | 93.08  | 472   | 1103  |
| RS       | 1232 | 715.082 | 91.534 | 463   | 1009  |
| SLG      | 1232 | 0.397   | 0.033  | 0.301 | 0.491 |
| W        | 1232 | 80.904  | 11.458 | 40    | 116   |

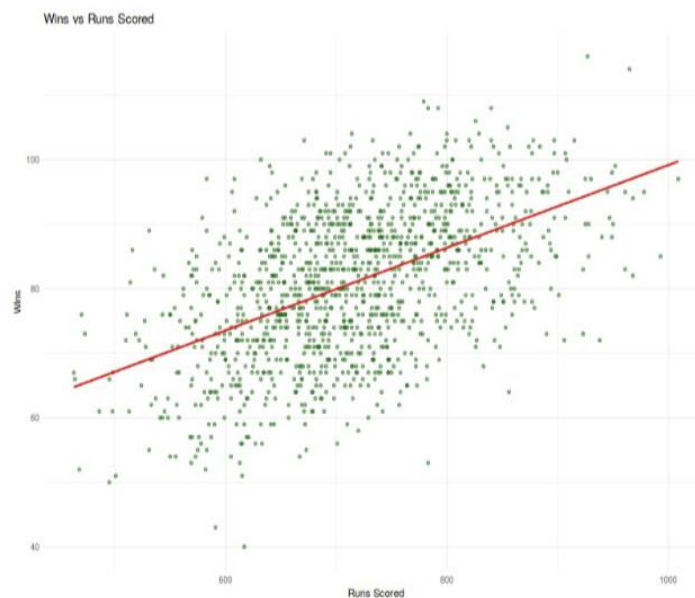
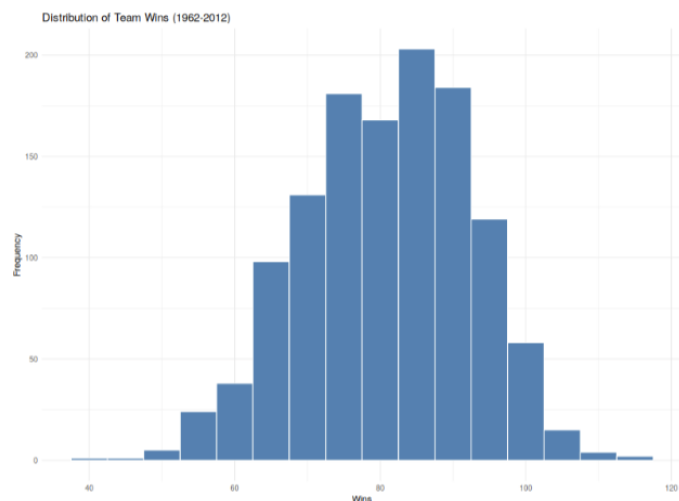
Team batting average is tightly clustered around 0.259 (SD = 0.013) while on-base percentage and slugging percentage show slightly more spread. Of the 1,232 team seasons, 244 (19.8%) resulted in a playoff berth. A correlation matrix becomes difficult to interpret with more than 5 variables so the correlation analysis here was limited to the 5 variables most relevant to team success which are wins, runs scored, runs allowed, on-base percentage, and slugging percentage.

| Variable | W      | RS    | RA     | OBP   | SLG   |
|----------|--------|-------|--------|-------|-------|
| W        | 1      | 0.512 | -0.532 | 0.482 | 0.401 |
| RS       | 0.512  | 1     | 0.38   | 0.9   | 0.919 |
| RA       | -0.532 | 0.38  | 1      | 0.326 | 0.437 |
| OBP      | 0.482  | 0.9   | 0.326  | 1     | 0.791 |
| SLG      | 0.401  | 0.919 | 0.437  | 0.791 | 1     |

Wins correlate positively with runs scored ( $r = 0.512$ ) and negatively with runs allowed ( $r = -0.532$ ), confirming the result that scoring more runs and allowing fewer runs both drive winning. On-base percentage and slugging percentage are both strongly associated with runs scored ( $r = 0.900$  and  $r = 0.919$  respectively) and the two offensive metrics are highly correlated with each other ( $r = 0.791$ ) which is expected since both are derived from similar, overlapping batting events.

The distribution of wins is approximately symmetric and bell-shaped, centered near 81 wins which is the midpoint of a 162-game season and is consistent with the zero-sum nature of

win-loss records within a league in a given year. The scatterplot of runs scored against wins shows a clear positive linear trend.



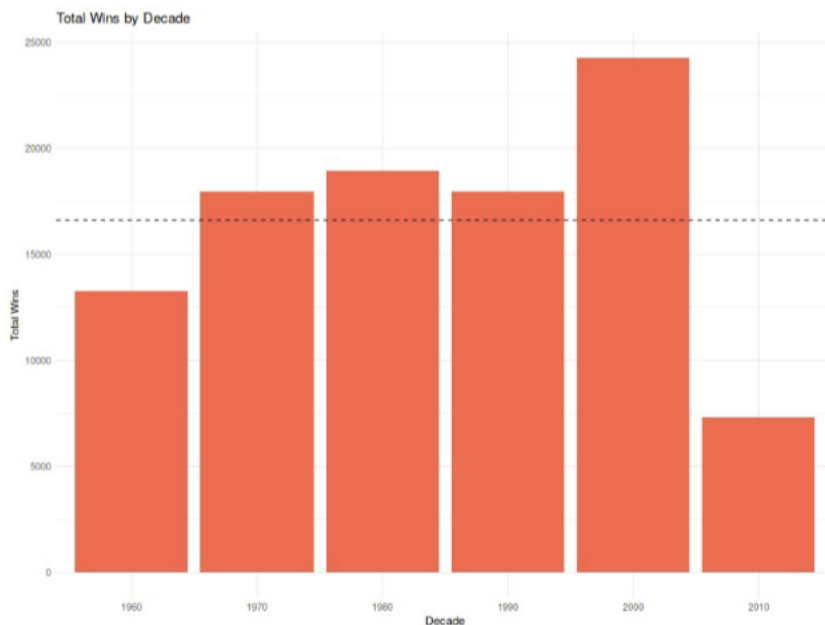
The visual diagrams suggest teams that score more runs tend to win more games, although the relationship is far from deterministic because pitching and defense (runs allowed) also play a substantial role in team success.

Wins by Decade

Using dplyr, the year variable was converted into a decade variable and total wins were summed within each decade. This produces a six-category frequency table with the 1960s, 1970s, 1980s, 1990s, 2000s, and 2010s.

| Decade | wins  | expected |
|--------|-------|----------|
| 1960   | 13267 | 16612.33 |
| 1970   | 17934 | 16612.33 |
| 1980   | 18926 | 16612.33 |
| 1990   | 17972 | 16612.33 |
| 2000   | 24286 | 16612.33 |
| 2010   | 7289  | 16612.33 |

Assuming the expected frequencies are equal across decades, a chi-square goodness-of-fit test was performed.  $H_0$  is that total wins are evenly distributed across decades (the claim under an equal frequency assumption) and  $H_1$  is that total wins are not evenly distributed across decades. At  $\alpha = 0.05$  with  $6 - 1 = 5$  degrees of freedom, the critical value is 11.070.



The computed chi-square statistic is 9,989.536 which far exceeds the critical value of 11.070 ( $p < 2.2e-16$ ) and the decision is to reject  $H_0$  since there is very strong evidence at  $\alpha = 0.05$  that

total wins are not evenly distributed across decades. The result should be interpreted with an important caveat which is that the decades are not equally represented in the underlying data. The dataset covers 1962 – 2012 so the 1960s and 2010s decades include only a partial set of years and fewer teams since expansion added several franchises over time, while the 2000s decade includes a complete 10 years with the largest number of teams. This uneven exposure rather than an actual shift in competitive parity is most likely the cause of the large chi-square statistic since total wins scale directly with the number of team seasons observed in each decade. Comparing the critical value method to the p-value method leads to an identical conclusion because the test statistic falls far into the rejection region defined by the critical value (11.070) and the p-value reported by R (2.2e-16) is far below  $\alpha = 0.05$ .

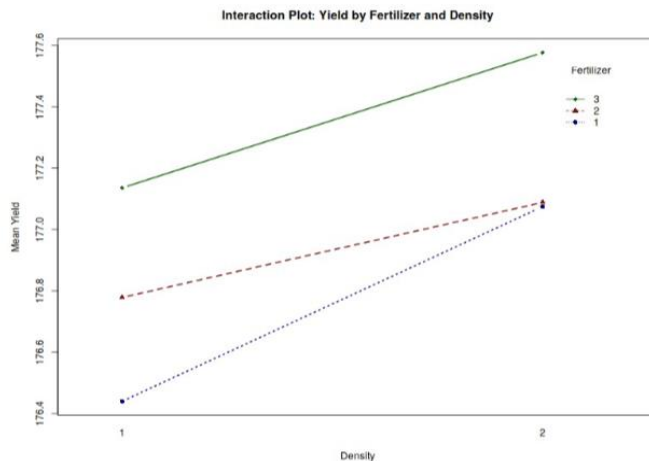
#### Crop Data

The `crop_data.csv` file was imported into R and contains 96 observations describing crop yield under two planting densities (1 and 2), three fertilizer treatments (1, 2, and 3) and four experimental blocks. Prior to analysis, density, fertilizer, and block were converted to R factors since each represents a categorical grouping variable. A two-way ANOVA was then fit with yield as the dependent variable and fertilizer and density as the independent variables. Three sets of hypotheses were tested. For the fertilizer main effect  $H_0$  mean yield is the same across all three fertilizer types and  $H_1$  mean yield differs for at least one fertilizer type. For the density main effect  $H_0$  mean yield is the same for both planting densities and  $H_1$  mean yield differs between the two densities. For the interaction effect  $H_0$  there is no interaction between fertilizer and density, and  $H_1$  fertilizer and density interact in their effect on yield.

| Fertilizer | Density | n | mean_yield | sd_yield |
|------------|---------|---|------------|----------|
|------------|---------|---|------------|----------|

|   |   |    |         |       |
|---|---|----|---------|-------|
| 2 | 1 | 16 | 176.778 | 0.541 |
| 2 | 2 | 16 | 177.088 | 0.58  |
| 3 | 1 | 16 | 177.136 | 0.51  |
| 3 | 2 | 16 | 177.577 | 0.615 |

| Source              | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|---------------------|----|--------|---------|---------|--------|
| fertilizer          | 2  | 6.068  | 3.034   | 9.001   | 0      |
| density             | 1  | 5.122  | 5.122   | 15.195  | 0      |
| fertilizer: density | 2  | 0.428  | 0.214   | 0.635   | 0.533  |
| Residuals           | 90 | 30.337 | 0.337   | NA      | NA     |



Both main effects are statistically significant at  $\alpha = 0.05$ . The fertilizer effect is significant ( $F = 9.001$ ,  $df = 2, 90$ ,  $p < 0.001$ ), exceeding the critical value of 3.098, so  $H_0$  (fertilizer) is rejected since mean yield differs across at least one pair of fertilizer treatments. The density effect is also significant since  $F = 15.195$ ,  $df = 1, 90$ ,  $p < 0.001$  which exceeds the critical value of 3.947 so  $H_0$  (density) is rejected because mean yield differs between the two planting densities with density level 2 producing higher average yield across all fertilizer groups. The fertilizer by density interaction is not significant since  $F = 0.635$ ,  $df = 2, 90$ , and  $p = 0.533$  which does not exceed the critical value of 3.098 so  $H_0$  (interaction) is not rejected. The interaction plot supports

the conclusion because the lines for the three fertilizer treatments are roughly parallel across both density levels which indicates the effect of density on yield is consistent regardless of which fertilizer is applied. There is a reason to believe that both fertilizer and density independently affect crop yield, but there is no evidence that the two factors interact. This means growers can consider each factor's effect separately when planning planting density and fertilizer strategy for this crop.

### **Conclusion:**

Across the textbook and independent datasets analyzed in this report the chi-square and ANOVA procedures produced a mix of significant and non-significant results that illustrate how these tests behave under different data conditions. The chi-square goodness of fit and independence tests was more likely to reject the null hypothesis when the sample sizes were large (airline performance, movie admissions, military rank all rejected  $H_0$  with very small p-values). This pattern reflects a known property of chi-square tests since statistical power increases with sample size. The one-way ANOVA problems (sodium content, company sales, per-pupil expenditures) all failed to reject the null hypothesis of equal means primarily because the within-group variability (standard deviations) was large relative to the differences between the group means. A numerically large sample mean difference does not guarantee statistical significance if the underlying data are highly variable or the samples are small. The two-way ANOVA problems told a deeper story and in the plant growth experiment, plant food had a strong significant effect on growth, while grow-light strength and food-by-light interactions did not reach significance. In the crop yield experiment, both fertilizer type and planting density affected yield on their own, but did not interact, meaning farmers can choose fertilizers and planting density independently from each other. The baseball chi-square goodness of fit test is a



cautionary example because it technically rejected the null hypothesis of equal wins across decades. The interpretation of this is that number of team seasons captured in each decade differs substantially rather than win totals meaningfully shifting from decade to decade.

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